BOE-704	Product Development	L T P 3 0 0	3 Credits

Pre-requisites: None

Course Objectives:

CO1	To understand basic concepts of product design, product features
CO2	Ability to coordinate multiple, interdisciplinary tasks in order to achieve a common objective.
CO3	Competence with a set of tools and methods for product design and development.
CO4	Reinforcement of specific knowledge from other courses through practice and reflection in an action-oriented setting.
CO5	Introduction about ROBUST DESIGN

Detailed Syllabus

MODULE I

Introduction: Need for IPPD, strategic importance of product development – integration of customer, designer, material supplier and process planner, Competitor and costumer – behavior analysis Understanding customer – promoting customer understanding – involve customer in development and managing requirements – Organization – process management and improvement – Plan and establish product specification.

MODULE II

Concept generation and concept selection: Activity of concept generation – Structured approaches – Five step Method: clarify – Search-Externally and internally – explore systematically – reflect on the solutions and processes – Concept selection – Integral part of PDD process-methodology – benefits. Investigation of customer needs –conceptualization – refinement – management of the industrial design process – technology driven products – user – driven products – assessing the quality of industrial design.

MODULE III

Product Architecture: Implications – Product change – variety – component standardization – Product performance – manufacturability Industrial design: Assessing the need for industrial design, impact – design process Integrate design process – assessing the quality of industrial design. ROBUST DESIGN – introduction, various steps in robust design.

Text Books:

1. Kari T.Ulrich and Steven D.Eppinger,"Product Design and Development", McGraw-Hill International Edns. 1999.

Course Outcomes: After the completion of the course the student will be able to:

CO1	Students will able to design some products for the given set of applications
CO2	knowledge gained through prototyping technology will help the student to make a prototype of a problem
CO3	Evaluate the role of design in product development, and the ability to address costs issues through better design decisions
CO4	Understand and discuss key concepts and principles concerning the activities and competencies involved in new product development.
CO5	Explain ROBUST DESIGN